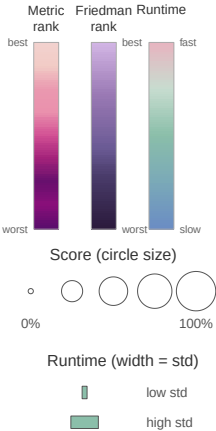


Overall method ranking Friedman rank–based composite

Method	Language	Category	Maximum produced Topology	Accuracy Metrics						Runtime
TSCAN	R	Cluster–based MST	Tree							5.74s
Slingshot	R	Cluster–based MST + principal curves	Tree							94.3s
PAGA + DPT	Python	Graph + diffusion pseudotime	Graph							0.49s
CellRank	Python	Probabilistic/Markov model	Directed Graph							4.06s
SCORPIUS	R	MDS + principal curve	Linear							6.61s
Monocle3	R	Principal graph	Tree / Graph							34.8s
ELPiGraph	Python	Principal graph	Tree / Graph							89.0s
VIA	Python	Graph + lazy random walk	Graph							14.2s



Accuracy metrics: circle colour encodes Friedman rank (light pink = rank 1/best, dark purple = rank 8/worst); circle size encodes normalised mean score across tasks (larger = higher score). Friedman rank column: filled square, p

Runtime column: rectangle colour encodes mean runtime (log scale) using the celestial palette ... soft pink = fast, periwinkle blue = slow; rectangle width encodes the standard deviation across the 6 biological tasks (wider rectang